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# Uber Ride Price Prediction

import pandas as pd
import numpy as np

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score

# Load dataset
data = pd.read_csv("uber.csv")

# Remove missing values
data = data.dropna()

# Remove invalid fare values
data = data[data["fare_amount"] > 0]

# Select input features
X = data[[
    "pickup_longitude",
    "pickup_latitude",
    "dropoff_longitude",
    "dropoff_latitude",
    "passenger_count"
]]

# Select output
y = data["fare_amount"]

# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
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# Linear Regression
lr = LinearRegression()
lr.fit(X_train, y_train)

y_pred_lr = lr.predict(X_test)

# Random Forest Regression
rf = RandomForestRegressor(
    n_estimators=100, random_state=42
)

rf.fit(X_train, y_train)

y_pred_rf = rf.predict(X_test)

# Evaluate Linear Regression
print("Linear Regression")
print("MAE:", mean_absolute_error(y_test, y_pred_lr))
print("MSE:", mean_squared_error(y_test, y_pred_lr))
print("R2 Score:", r2_score(y_test, y_pred_lr))

# Evaluate Random Forest
print("\nRandom Forest Regression")
print("MAE:", mean_absolute_error(y_test, y_pred_rf))
print("MSE:", mean_squared_error(y_test, y_pred_rf))
print("R2 Score:", r2_score(y_test, y_pred_rf))
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